

CLARION TACKLE CO. · TAMPA BAY, FLORIDA

A GUIDE ON GUIDES.

What the rings and frames on your rod actually do, what the materials really change, and how to tell which setup is right for the way you fish.

Written by a builder, not a catalog. No brand is paying for placement in here — where I name a product it's because it's worth naming, and where I think something is oversold I say that too.

Every specification in this booklet is taken from the manufacturer's own published tables and is reproduced as published. Where a figure is a manufacturer's performance claim rather than a measured property, it is labelled as one. Where I'm giving you my opinion, I say "I".

START HERE

EVERYBODY ARGUES ABOUT BLANKS. THE GUIDES ARE WHAT YOU FEEL.

Ask ten anglers what makes a rod good and nine will start talking about the blank. Modulus, taper, where it's rolled. Fair enough — the blank is the engine.

But the guides are the only place your line ever touches the rod. Every foot of every cast runs through them. Every headshake, every bottom tick, every moment of a fish turning gets transmitted through them and into the blank before it reaches your hand. Get the guides wrong on a great blank and you have a mediocre rod. Get them right on a good blank and you have something that feels like it's reading the water for you.

They're also where most factory rods quietly save money, because it's the change a buyer in a store is least likely to notice.

**THE BLANK DECIDES WHAT THE ROD CAN DO.
THE GUIDES DECIDE WHAT YOU CAN FEEL.**



THE ONLY PLACE YOUR LINE EVER TOUCHES THE ROD.

HOW TO USE THIS

Four sections. What a guide actually does, then the ring, then the frame, then the train — the layout of guides along the blank, which matters more than either material choice and gets the least attention.

You don't need to memorize any of it. When we build your rod, choosing this is my job. But you're spending real money on a tool you'll fish for fifteen years, and you should know what you're paying for and what you're not.

WHAT A GUIDE ACTUALLY DOES.



A SINGLE-FOOT GUIDE ON A CLARION BUILD.

Three jobs, and they pull against each other. Every design decision in this booklet is a trade between them.

Ring, frame, foot, and placement. Four choices, made twenty or thirty times down a blank, and together they decide more about how a rod fishes than the sticker on the butt section does.

JOB ONE

CONTROL THE LINE

Line doesn't leave a reel in a straight line. Off a spinning reel it comes off the spool in expanding loops; off a casting reel it wanders. The first guides have to gather that mess and get it running straight down the blank without slapping the sides. Do it badly and you get friction, lost distance, and wind knots.

JOB TWO

TRANSFER THE LOAD

When a fish pulls, that force travels up the line, into the guides, and into the blank. Where the guides sit determines how the rod bends — whether the blank loads through a smooth curve or hinges at a flat spot. That's why guide placement is measured and tested rather than eyeballed, and it's why two rods on identical blanks can fight completely differently.

JOB THREE

SURVIVE THE FRICTION

Line moving under load generates heat. Braid is essentially a fine abrasive — it will cut a soft ring like a saw over a season. The ring has to be hard enough not to groove and conductive enough to shed heat during a long run, because a hot ring is what melts a leader knot at the worst possible moment.

THE RING.

The ring is the insert your line rides on — the part that determines friction, heat, and how long before your guides develop grooves. It's also where the price ladder lives.

Roughly, harder means smoother and longer-lasting. Higher thermal conductivity means better at dumping heat on a long run. Thinner and lighter means less mass on the blank. No single material wins all three, which is the entire reason there's a choice to make.

These are Fuji's own published numbers for their own rings. Read them carefully, because they don't say what most people assume they say.

PROPERTY	TORZITE	SiC	SIN II	ALCONITE	"O" RING
Vickers hardness (GPa)	13-15	22-24	13-15	13-15	12-14
Specific gravity	3.3	3.2	3.2	4.2	3.8
Thermal conductivity (W/m·K)	27	60	20	11	13
Bending strength, 3PB (MPa)	1,020	540	610	440	300

Source: Fuji Kogyo published guide ring physical properties. For scale, Fuji lists diamond at 60 GPa hardness and 138.1 W/m·K. "O" Ring is aluminium oxide; Alconite is reinforced aluminium oxide.

TWO THINGS IN THAT TABLE WORTH STOPPING ON

Torzite is not harder than SiC. It's not even close. SiC runs 22–24 GPa; Torzite runs 13–15 — the same band as Alconite. The most expensive ring Fuji makes is softer than the one below it. Plenty of what gets written about Torzite has this backwards, which is why I'm quoting the manufacturer's own table rather than paraphrasing somebody's blog.

SiC sheds heat more than twice as well as Torzite — 60 against 27 — and better than five times as well as Alconite. If your fish make long, hard, drag-screaming runs, the premium ring is not the right ring.

WHY TORZITE COSTS MORE.

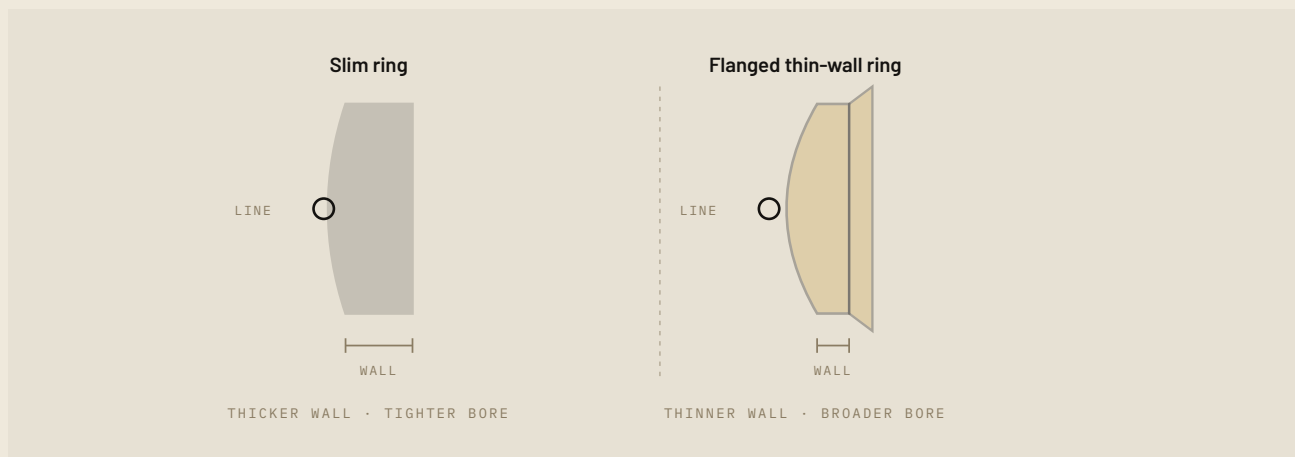
If Torzite is softer than SiC, worse at heat than SiC, and — at 3.3 against 3.2 specific gravity — very slightly *denser* than SiC, why is it the expensive one?

One number in that table answers it.

THE WHOLE STORY

BENDING STRENGTH: 1,020 MPA VS 540

Torzite is nearly **twice as strong in bending** as SiC. That doesn't make it tougher against your line — it makes it possible to build the ring with a much thinner wall without it cracking. Torzite isn't a lighter material. It's a material strong enough to use less of.



Cut through one side of the ring, where your line rides. Fuji sells Torzite as a *flanged* ring and SiC as a *slim* ring. Illustrative, not to scale.

A thinner wall means a larger opening inside the same outside dimension — so a Torzite ring passes the same line as a physically bigger SiC ring, and you can often drop a size. Less ceramic, smaller frame, less mass out toward the tip, where the blank recovers faster for it. On a light finesse rod that's a difference you'll feel in an afternoon.

Where I'd spend it, and where I wouldn't. In Fuji's current catalogue Torzite appears only on titanium frames, so you're buying both. On a finesse spinning rod where every gram at the tip counts, it's worth the money. On a heavy offshore rod facing long runs and hot drags, I'd take SiC for the heat handling and put the difference into a better blank.

Every number in this section comes from Fuji's published specification tables. Where I've made a judgement rather than cited a figure, I've said so in plain language.

THE FRAME.

The ring gets the attention. The frame is what holds it, and it decides durability, weight, and whether your guides are still straight after a season in the boat. Fuji publishes numbers here too, and they make the case better than any sales pitch.

PROPERTY	HIGH-TENSILE TITANIUM	PURE TITANIUM	STAINLESS STEEL
Vickers hardness (GPa)	4.04	0.98	1.81
Specific gravity	5.03	4.51	7.93
Tensile strength (MPa)	1,474.9	343	519.4
Yield strength (MPa)	1,450.4	215.6	205.8
Corrosion resistance	★★★★★	★★★★★	★★

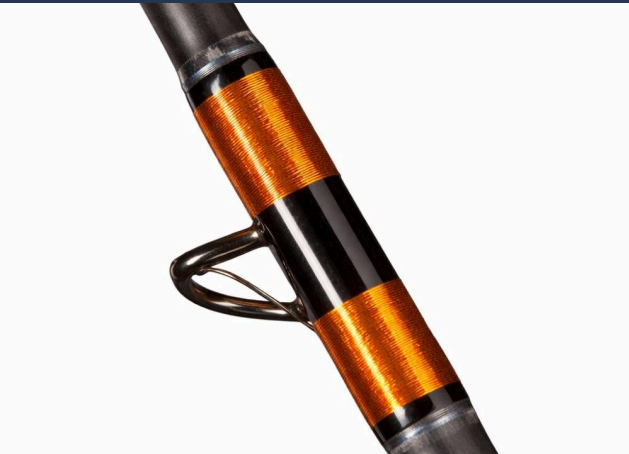
Source: Fuji Kogyo published guide frame physical properties. Star ratings are Fuji's own.

Weight. Specific gravity 5.03 against 7.93 — titanium is about a third lighter than stainless for the same volume of metal. **Corrosion:** Fuji rates both titanium grades five stars against stainless at two, and over years in salt that gap is real.

Why a titanium guide springs back. Yield strength is where metal stops bending temporarily and starts bending permanently. High-tensile titanium yields at 1,450 MPa; stainless at 206 — roughly **seven times the stress before it takes a set**. Step on a stainless guide and it stays folded over. Step on a titanium one and it usually springs back. On a rod that lives in a boat that beats the weight saving, and it's the other half of the Torzite price tag.



ONE GUIDE POSITION: FRAME, RING, FOOT, WRAP.



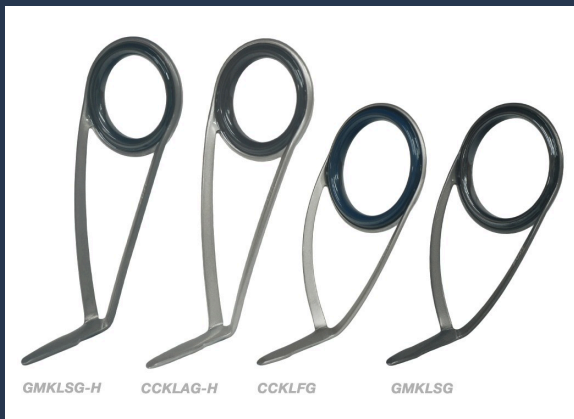
FOUR DECISIONS, REPEATED DOWN THE BLANK.

FEET, AND THE SHAPE OF THE FRAME.

Two more decisions get made at every guide position, and both change how the rod behaves.



DOUBLE FOOT • FUJI KW SERIES, CASTING AND SPINNING. TWO FEET, AND A SLOPED BRIDGE THAT SHEDS A LOOSE LOOP.



SINGLE FOOT • FUJI KL REDUCTION GUIDES.

HOW MANY FEET

Double foot is wrapped at both ends — stronger, and the right call anywhere the guide takes a beating: strippers, offshore rods, surf rods. The trade is weight, more thread, and a slightly stiffer spot on the blank at each wrap.

Single foot is wrapped once — lighter, less thread, and it lets the blank flex more naturally. Standard for running guides on freshwater and inshore rods.

Most well-built rods use both. Double foot down low where the load lives, single foot up top where weight matters. That's not a compromise — it's the right answer in each place.

Look at the frames themselves and you can see the tangle-free thinking: the ring leans forward on its bridge so a loose loop of braid slides off instead of catching. A small change to a piece of bent wire that solves a problem every angler who fishes braid on spinning gear knows by heart.

THE TRAIN.

This is the part almost nobody asks about and the part that changes the rod most.

"Guide train" just means the arrangement — how many guides, what sizes, and where they sit. Put the finest rings made on a rod, lay them out badly, and it'll cast worse than a factory stick.

THE OLD WAY

BIG GUIDES, SPREAD FAR APART

Conventional layouts used a few large guides, on the theory that a big opening lets the line's expanding loops pass without touching. It worked reasonably well with monofilament. Then braid arrived — limper, thinner, and far more willing to slap around and tie itself in knots — and the old logic started costing distance and creating wind knots.

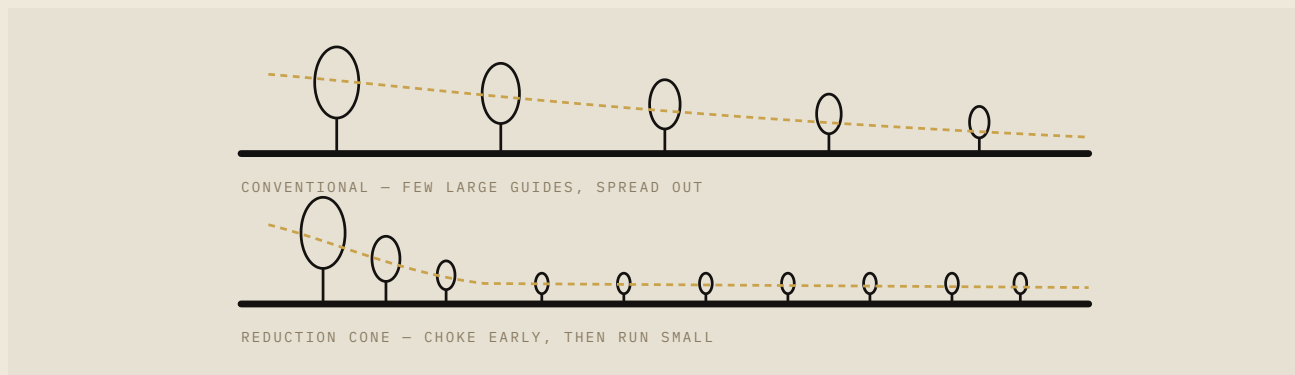
THE CURRENT THINKING

CHOKE IT DOWN EARLY, THEN RUN SMALL

Fuji's New Guide Concept, and the KR Concept that followed it for braid, flipped the approach. Instead of a few big guides, you use a short **reduction cone** — a small group of guides near the reel that quickly chokes the line down from the wide loops coming off the spool into a controlled, straight path. After that the line runs through a series of small, light guides sitting close to the blank. Fuji publishes placement charts for it; the exact count and spacing depend on rod length, reel size and line.

Fuji names the parts after the job: the KL guides on the previous page are sold as *reduction* guides — that's the cone. Their K-Series frames carry a "Tangle Free" designation, angled so a loose loop sheds rather than catching. The claimed payoff is more distance, fewer tangles, less weight and a more even bend. In practice the tangle reduction with braid is the part that's hardest to argue with.

Illustrative. Control the line early and you can run smaller the rest of the way.



THE MICROWAVE SYSTEM.

American Tackle took the same problem — the loose, expanding loop of line coming off the reel — and attacked it differently.

Instead of choking the line down over three guides, the MicroWave system does it at the **first guide**, using a stripper with two rings: a larger outer ring that catches the wide loop immediately, and a smaller inner ring that the line settles into. The line gets gathered and straightened right where it leaves the spool, before it has any room to misbehave.

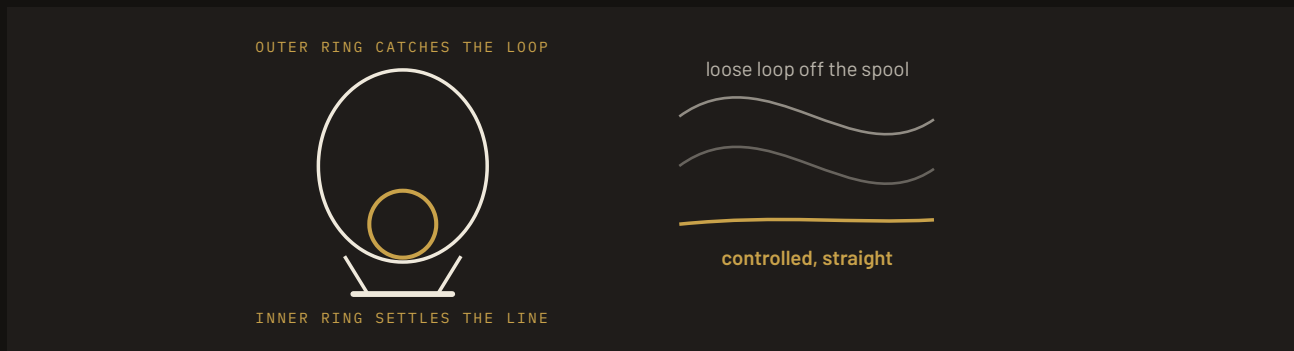


Diagram: my own illustration of the principle, not a product drawing.

WHAT THEY CLAIM

- More casting distance and accuracy
- Fewer guides needed overall
- Wind knots largely eliminated
- Better balance from less guide weight

MY READ ON IT

It genuinely works, and it's a clever piece of engineering. Whether it beats a well-laid-out KR train is the argument builders have been having for a decade, and honestly the answer depends more on the rod and the reel than on the system.

Where I'd actually use it. It shines on spinning rods with braid where wind knots are the problem you're trying to solve, and it looks distinctive — some people love that, some don't want their rod to look different. It's not a magic distance machine, and anyone who tells you a guide system will add twenty yards to your cast is selling something. What it does reliably is make a rod more forgiving — and if you've spent any part of your life picking knots out at the stripper, that's not small.

SO WHAT SHOULD BE ON **YOUR** ROD?

Here's how I'd actually decide, stated plainly.

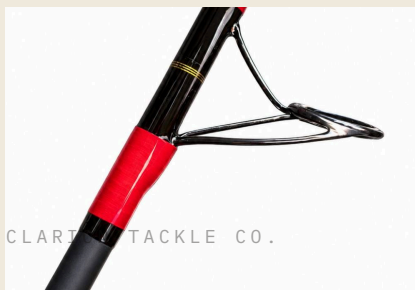
IF YOU'RE...	RING	FRAME & TRAIN
Fishing mono or fluoro in fresh water on a budget	Alconite	Stainless, conventional or KR train. Do not overspend here — put it in the blank.
Throwing braid inshore, all day, every trip	Alconite or SiC	Stainless or titanium, KR train. This is the sweet spot for most Florida fishing.
Finesse fishing where feel is everything	Torzite	Titanium (it comes no other way), KR train, single foot runners. Tip weight is the whole game.
Offshore, heavy drag, long runs	SiC	Titanium, double foot low. Heat handling beats weight saving here.
Fighting wind knots on spinning braid	Any	MicroWave stripper, or a properly executed KR reduction cone.
Fishing surf or anywhere gear gets abused	Alconite or SiC	Double foot, titanium if it lives in salt. Toughness over everything.

THREE THINGS WORTH KNOWING

The most expensive ring is not always the right ring. By Fuji's own numbers SiC is harder than Torzite and sheds heat more than twice as well. If your fish run hard, that beats forty percent off the ring weight every time.

Layout beats material. A well-laid-out Alconite train on a good blank will out-fish a badly laid-out Torzite train every day of the week. If you only care about one thing on this list, care about the train.

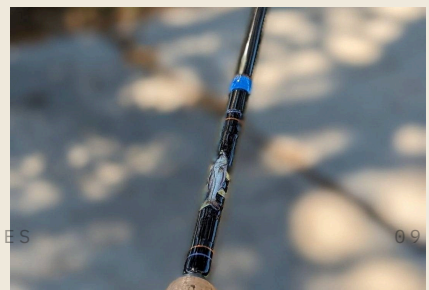
Guides are the cheapest place to buy real performance and the first place a factory cuts. A production rod hitting a price point will use heavier frames and cheaper rings than the blank deserves, because that's the part you can't see from across the store.



CLARITY TACKLE CO.



A GUIDE ON GUIDES



09

LAST PAGE

NONE OF THIS IS YOUR JOB.

You just read nine pages on guides. Here's the honest truth about why: so that when I tell you what I'm putting on your rod, you know whether I'm making a real decision or just selling you the most expensive box on the shelf.

When I build a rod, the guide train comes out of your answers — where you fish, what you're chasing, what line you run, how you cast. Not out of a catalog, and not out of whatever I have the most of on the bench.

**IF I WON'T FISH IT,
I WON'T BUILD IT.**



NEXT STEP

TELL ME WHAT YOU FISH.

Every Clarion build starts with a conversation about where you fish, what you're after, and what your last rod got wrong. Takes about three minutes. No account, no obligation, no deposit.

clarionrods.com · hello@clarionrods.com · [@clarionrods](https://www.instagram.com/clarionrods)

Builds start at \$600 · Current lead time 12 weeks · \$50 flat shipping

Sources: Fuji Kogyo 2026 product catalogue (guide ring physical properties, ring and frame specifications), plus published technical claims from Fuji Tackle and American Tackle for their own products and rod-building references from Mud Hole Custom Tackle and RodHouse. Manufacturer performance claims are identified as such throughout. Fuji, Torzite, Alconite, SiC, K-Series and KR Concept are trademarks of Fuji Kogyo Co., Ltd. MicroWave is a trademark of American Tackle Company. Clarion Tackle Co. is not affiliated with, sponsored by, or endorsed by either company.